🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.1

Fraction Division Problem Solving

Lesson 6-11 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes
Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve real-world problems by writing and solving fraction division equations.

Language Objective: I can explain my reasoning using the words model, equation, solution, and reasonableness.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Fraction Division Problem Solving | Model | Equation | Solution | Reasonableness | Inverse operations

👆 *Tap any word to see what it means and a picture.*

1. Name the total and the size of each group, then divide the ________ by the ________.

2. A drawing or diagram that shows a math situation is a ________.

3. A math sentence that says two amounts are equal, with an = sign, is an ________.

4. The value that makes an equation true is the ________.

5. Checking whether an answer makes sense for the problem is checking its ________.

6 Operations that undo each other, like multiplication and division, are

____________________ .

Write About the Math

How does setting up a fraction division equation help solve this real-world problem?

¿Cómo ayuda armar una ecuación de división de fracciones a resolver este problema del mundo real?

Say your answer to a partner first. Then write it, using at least two of these words:

- [ ] **Fraction Division Problem Solving** *Resolución de problemas de división de fracciones*
- [ ] **Model** *Modelo*
- [ ] **Equation** *Ecuación*
- [ ] **Solution** *Solución*
- [ ] **Reasonableness** *Razonabilidad*

Model: *The total amount is the dividend; the size of each test is the divisor.* — Students identify 2/3 as the total being split (dividend) and 1/6 as the group size (divisor), writing 2/3 ÷ 1/6.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The chemist can run ____ tests because 5/6 ÷ 1/12 = ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

____

____

____

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Fraction Division Problem Solving
2. **Notes 2:** Model
3. **Notes 3:** Equation
4. **Notes 4:** Solution
5. **Notes 5:** Reasonableness
6. **Notes 6:** Inverse operations
7. **Watch for this mistake:** *A common mistake in Fraction Division Problem Solving is reversing which fraction is the dividend and which is the divisor when setting up the equation from a word problem. For example: a pitcher holds 3/4 cup of punch, and each serving is 1/8 cup — how many servings? The correct setup is total ÷ group size = 3/4 ÷ 1/8 = 3/4 × 8/1 = 24/4 = 6 servings. A student who reverses the setup calculates 1/8 ÷ 3/4 = 1/8 × 4/3 = 4/24 = 1/6, an answer smaller than 1 that can't possibly be a serving count. Before you submit, check: does my answer make sense as a NUMBER OF GROUPS? If the setup gives a fraction less than 1 for a 'how many' question, you likely swapped the dividend and divisor.*
8. **Try It 1:** A. 5 — *The total distance (5/6 mile) is the dividend and the checkpoint spacing (1/6 mile) is the divisor: 5/6 ÷ 1/6 = 5/6 × 6/1 = 30/6 = 5 checkpoints.*
9. **Try It 2:** A. 3 — *The total (9/10 liter) is the dividend and each vial (3/10 liter) is the divisor: 9/10 ÷ 3/10 = 9/10 × 10/3 = 90/30 = 3 vials, so the code is 3.*